Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/26/2024
PI Name:	Hein, Travis Ph.D.		
Project Title:	Temporal Impact of Simulated Microgravity on Ocular Vascular Hydrodynamics		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	08/01/2019	**End Date:**	07/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2025 per NSSC information (Ed., 7/3/24) NOTE: End date changed to 7/31/2024 per F. Hernandez/ARC (Ed., 6/21/23) NOTE: End date changed to 7/31/2023 per F. Hernandez/ARC and NSSC information (Ed., 6/20/22) NOTE: End date changed to 7/31/2022 per F. Hernandez/ARC (Ed., 3/17/2020)		
Key Personnel Changes/Previous PI:	Our Co-Investigator, Dr. Robert Rosa, left our institution during the first few months of the current reporting period to join an ophthalmology clinic that does not allow research time, so he was removed from this project. One postdoc, with the ophthalmology experience as did Dr. Rosa to perform the live animal imaging of OCT for this project, was hired in the middle of this reporting period		
COI Name (Institution):	Zawieja, David Ph.D. (Texas A&M University System) Tharakan, Binu Ph.D. (Morehouse School of Medicine) Macias, Brandon Ph.D. (NASA Johnson Space Center) Lee, Stuart Ph.D. (KBR/NASA Johnson Space Center) Laurie, Steven Ph.D. (KBR/NASA Johnson Space Center)		
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Performance Goal No.:	
Performance Goal Text:	
Task Description:	Spaceflight associated neuro-ocular syndrome (SANS) is reported to affect about 40% of astronauts completing long-duration spaceflights and is characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, retinal nerve fiber layer thickening, or choroidal folds. Although prolonged exposure to the headward fluid shift occurring in microgravity is regarded as the primary instigating factor, the pathophysiology of SANS remains unclear. An additional gap in our knowledge is whether ground-based models of microgravity can simulate SANS. Therefore, ground-based studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we propose to focus on potential changes directly to the ocular vasculature caused by simulated microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen is dependent on retinal blood flow. Recent studies have reported elevated central retinal artery blood flow velocity during long-term spaceflight. A sustained increase in retinal blood flow during microgravity exposure may contribute to optic disc edema because of the elevated transcapillary pressure. In addition, a microgravity induced headward venous fluid shift, resulting in elevated cephalad venous pressures, may impede removal of ocular interstitial fluid, resulting in optic disc edema and contribute to SANS. The development of optic disc edema also may be exacerbated by disruption of the blood-retinal barrier and/or reduction in ocular lymph flow. A potential molecular mechanism promoting these vascular changes in the eye is an inflammatory response related to edema formation. Elevated tissue fluid increases the diffusion distance for oxygen and other nutrients. Edema also compromises diffusion mediated removal of toxic cellular metabolites. However, so far there has been no systematic analysis of ocular vascular changes in microgravity that may promote ocular edema and/or other characteristics of SANS. To address these information gaps, we have assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics). The objective of this application is to determine whether the hindlimb unloading (HU) model of simulated microgravity causes a time-dependent change in structure and function of the ocular vasculature at the level of feed arteries, venous exchange vessels and lymph vessels, and whether sex differences influence these responses. We will perform in vivo measures of vascular function (retinal artery blood flow velocity, retinal arteriolar and venular diameters, and retinal venular permeability) and in vitro studies of isolated vessel structure and function (vessel histology, arterial vasomotor regulation, and lymphatic transport characteristics) in male and female mice after 2- and 4-wk of HU. The central hypothesis of this proposal is that simulated microgravity causes temporal changes in structure/function of the ocular vasculature and subsequent alterations in ocular hydrodynamics leading to SANS. We will pursue 3 specific aims: (1) Determine the temporal impact of simulated microgravity on ocular artery structure/function; (2) Determine the temporal impact of simulated microgravity on ocular vein structure/function; (3) Determine the temporal impact of simulated microgravity on ocular lymphatic structure/function. Our team approach to accomplish these studies will provide a novel comprehensive evaluation of the temporal impact of simulated microgravity on ocular hydrodynamics in mice, and correlation of these findings with SANS. Outcomes will help identify the strengths and limitations of a murine HU model for SANS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our ground-based studies in this project will address whether relatively short- and long-term simulated microgravity using the hindlimb unloading model in mice can simulate signs and symptoms of spaceflight associated neuro-ocular syndrome (SANS), such as ocular edema and thickening of the retinal tissue. We will also determine whether simulated microgravity causes a time-dependent change in structure and function of the blood vessels in the eye, and whether ocular vascular changes correlate with SANS. Results from this project are expected to provide the first insight into the impact of simulated microgravity on vascular hydrodynamics in the eye. The expected findings will likely also provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma and ocular hypertension.
Task Progress:	• For Aim 1A, we completed the 2-wk and 4-wk hindlimb unloading (HU) in male and female mice data collection for the optical coherence tomography (OCT) retinal imaging. The total retinal thickness (TRT) measurements from the internal limiting membrane (ILM) to the retinal pigment epithelium (RPE) in the retinal images will require the no-cost extension period to complete. • For Aim 2A, the proposed studies to assess retinal microvascular permeability in HU and Sham control mice were initiated recently. The 4-wk HU and 4-wk Sham male mice are currently undergoing intravital microscopic imaging to determine blood-retinal barrier integrity/permeability. Mice are receiving FITC-dextran (10-kDA) as a fluorescent tracer for intravital imaging of the retinal microcirculation. We will be able to complete the intravital imaging data collection and analysis in male and female mice during the no-cost extension period. • For Aim 2B, the 4-wk HU and 4-wk Sham control studies of male mice have been completed and retinal tissue was collected to perform mRNA analysis of tight junction proteins. The results demonstrate a significant reduction in the mRNA expression of some inter-endothelial tight junction proteins (e.g., zonula occludens-1 [ZO-1]) in the retina as well as the brain from 4-wk HU mice (n=9) compared to 4-wk Sham control mice (n=12). In addition, 4-wk HU and control study for immunohistochemical localization of tight junction proteins in the retinal tissue was also completed. We are currently imaging and analyzing these data. We will be able to complete this study during the no-cost extension period. These data were presented at the NASA Human Research Program (HRP) Investigators' Workshop (IWS) meeting this past reporting year. • For Aim 2C, the proposed studies to determine mediators of inflammation after 4-wk was completed. Retinal and brain tissues were collected from 4-wk Sham control and 4-wk HU groups. The tissues were processed for gene expression analysis (by reverse transcription-polymerase chain reaction / RT-PCR) of the inflammasome component, NLRP3. Furthermore, expression of caspase-1, downstream of the NLRP3 inflammasome, is also being analyzed. We will be able to complete the molecular analysis for Aim 2C during the no-cost extension period.
Bibliography Type:	Description: (Last Updated: 07/07/2026)

Abstracts for Journals and Proceedings	Waithe OY, Muthusamy S, Laurie S, Macias BR, Zawieja DC, Hein TW, Tharakan B. "Effect of simulated microgravity on blood-brain and blood-retinal barrier tight junction proteins in a mouse model of hindlimb unloading." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024